

Samay Pandey

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Professional Summary

Systems and Machine Learning Engineer specializing in Computer Vision, backend systems, and security-aware infrastructure. Strong expertise in C++, Linux-based development, and performance optimization with experience in production ML pipelines, cloud deployment, and secure backend systems.

Technical Skills

- **Languages:** C++, Python, SQL, Bash
- **Machine Learning:** PyTorch, TensorFlow, YOLOv8/YOLOv11, OpenCV, multimodal systems
- **Backend:** FastAPI, Flask, REST APIs, database design, model serving
- **Systems & Cloud:** Linux (2.5+ years), Docker, Azure, AWS, GCP, Git
- **Security:** OWASP Top 10, Web Security, Authentication & Access Control, Burp Suite, Wireshark
- **Core Strengths:** System Optimization, ML Deployment, Distributed Systems, Secure Backend Design

Experience

BigShorts — Software Developer Intern

Dec 2024 – Mar 2025

- Built and deployed a production-scale multimodal ML pipeline for hierarchical content classification using YOLO models
- Engineered a Linux-based Azure VM training system optimized for CPU workloads, improving iteration speed by 20% and reducing cloud costs by 20%
- Optimized backend data workflows and modified Ultralytics framework, reducing data acquisition time by 60%

IETE-SF KCCEMSR — Technical Head

Jul 2025 – Present

- Led a 10+ member technical team across software and hardware domains
- Improved execution efficiency by 30% through structured planning and technical reviews

Projects

SmartNav – IoT Smart Wheelchair System

GitHub: <https://github.com/ChampionSamay1644/Smart-Wheelchair>

- Built a Linux-based voice-controlled navigation system using Whisper Tiny, achieving 6s end-to-end speech-to-action latency under real-time conditions
- Engineered Bluetooth-based control with sub-second response time and integrated WebSocket-based cross-platform control interface
- Implemented real-time camera streaming at 30 FPS (HD) for navigation feedback
- **Awards:** Aavishkar Finalist 2025 (University of Mumbai Research Convention); 2nd Prize – SCOE AVISHKAR 2026 (National-Level Project Competition); 5th Rank State Level – Colloquium'26 (SFIT)

Data Dash – Cross-Platform File Sharing App

Live: <https://datadashshare.vercel.app/>

- Built a TCP-based file transfer system supporting Android, Windows, macOS, and Linux, validated across all platforms
- Achieved 10–15% faster transfer speeds compared to Nearby Share and ShareIt through optimized socket-level data streaming
- Ensured 99%+ data integrity with minimal packet loss using reliable TCP-based transmission under concurrent usage

Achievements

- Aavishkar Finalist 2025 – University of Mumbai Research Convention
- 2nd Prize – SCOE AVISHKAR 2026 (National-Level Project Competition, Saraswati College of Engineering)
- 5th Rank State Level – Colloquium'26, SFIT
- 5th Rank Nationally – RAIT-CTF Finals
- 3rd Place – Techgyanathan Cybersecurity Competition

Security Experience

- Applied exploitation techniques across web security, forensics, OSINT, reverse engineering, and network traffic analysis (Rank 5 – RAIT Finals)
- Secured 3rd place at Techgyanathan cybersecurity competition (Vasantdada Patil College)
- Performed vulnerability testing on OWASP Juice Shop, identifying injection and access control flaws

Education

B.E. Computer Engineering, K. C. College of Engineering, Thane

CGPA: 8.6/10 (till Sem 7)

Relevant Coursework: DSA, Operating Systems, DBMS, Computer Networks